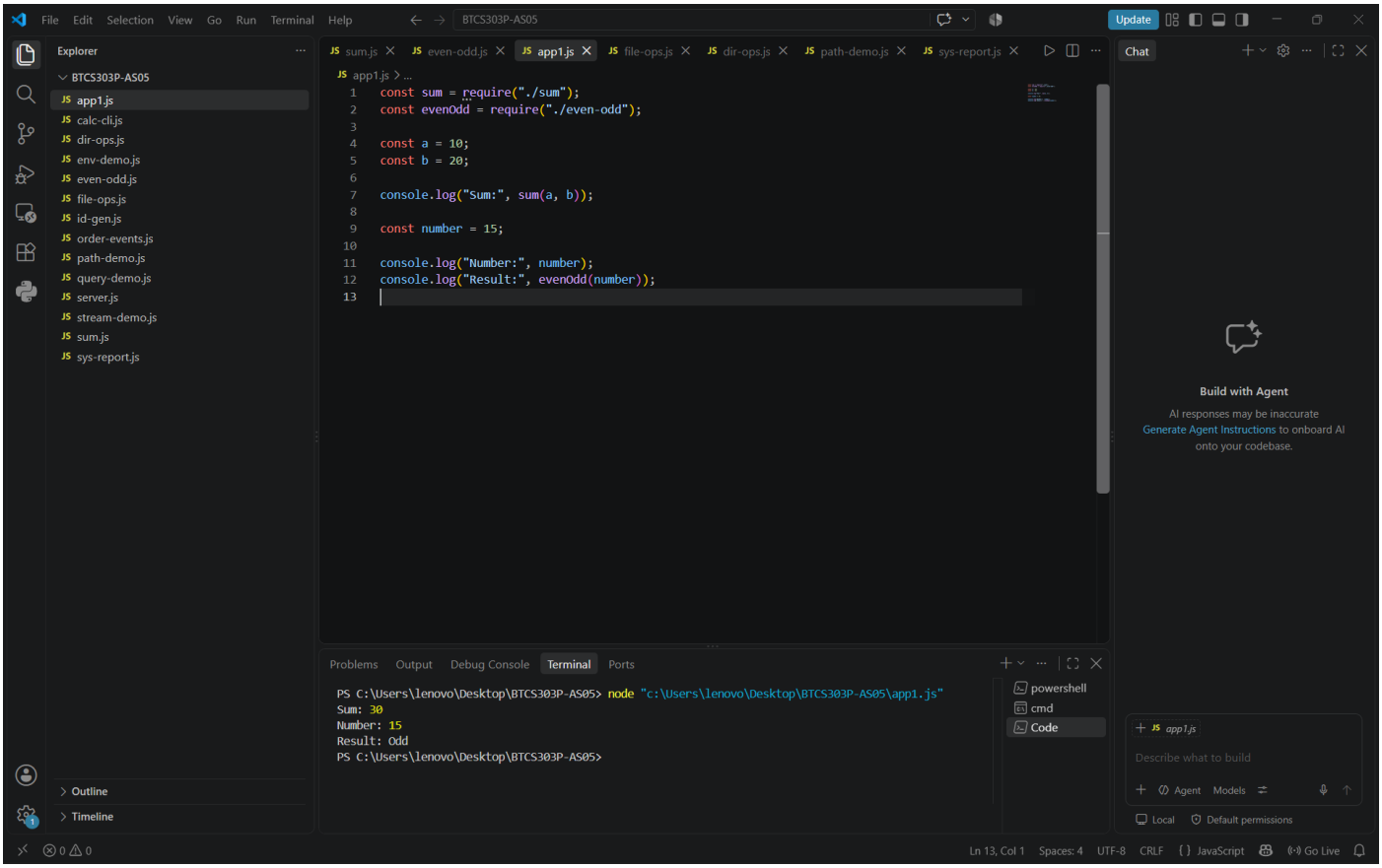
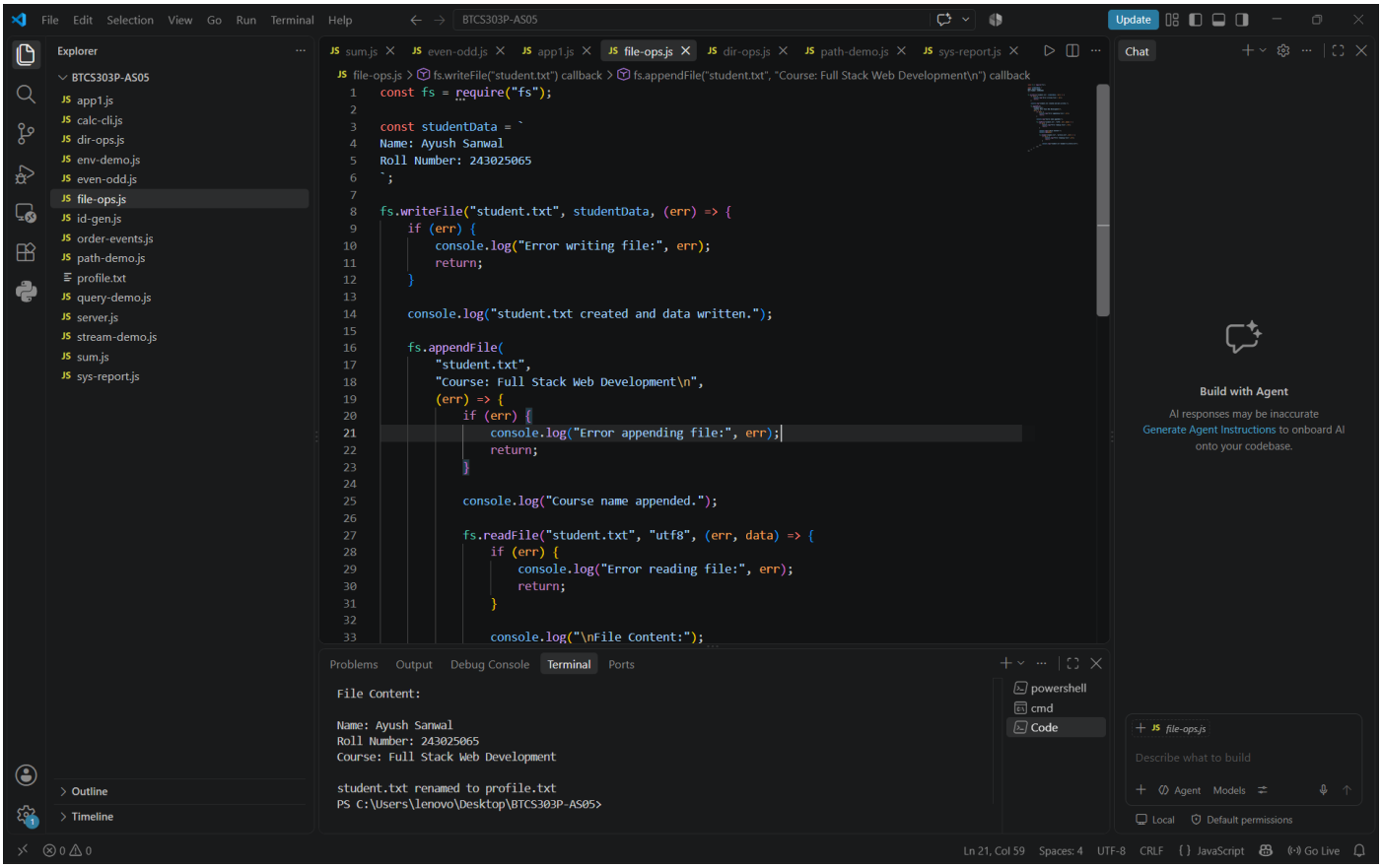
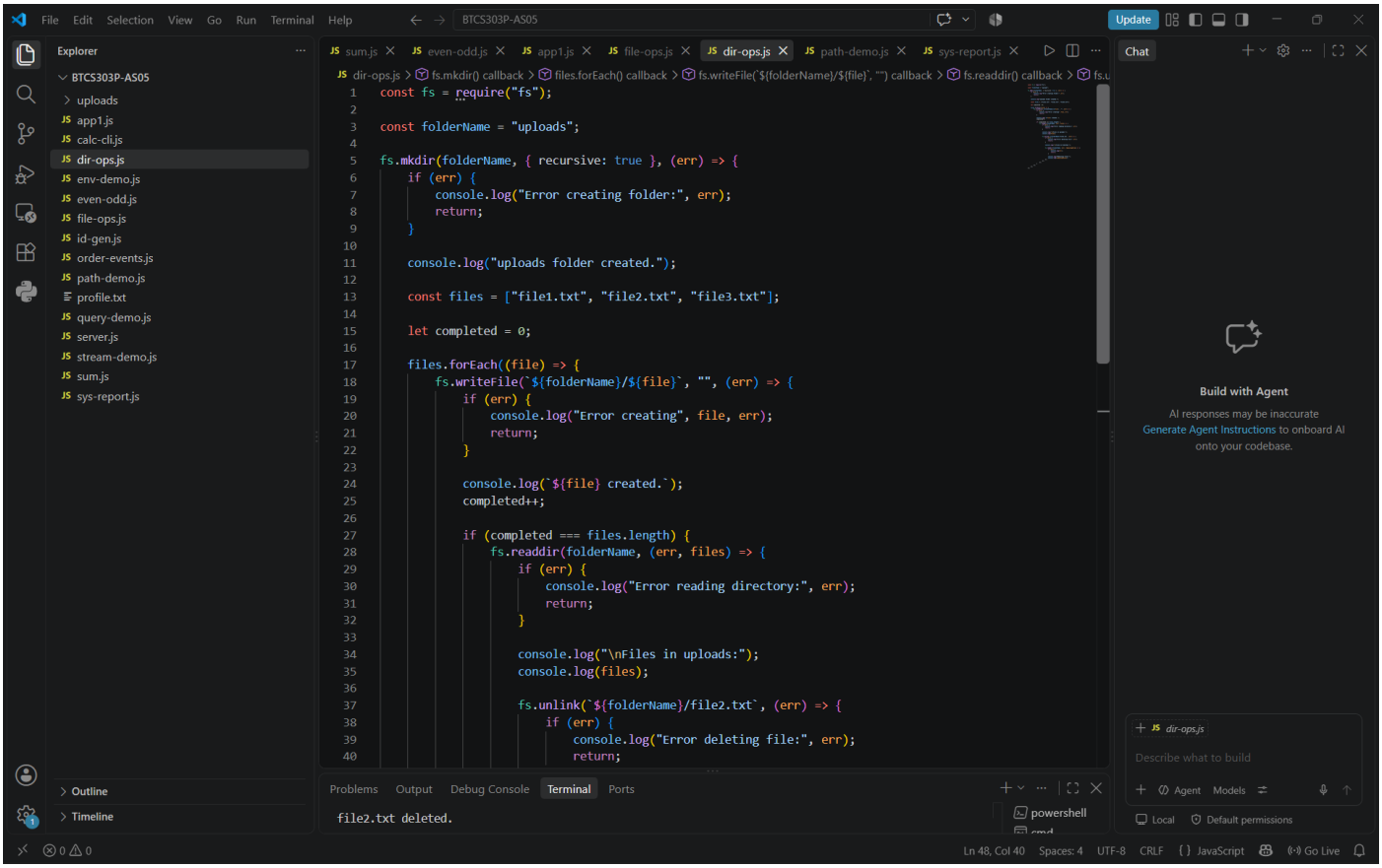
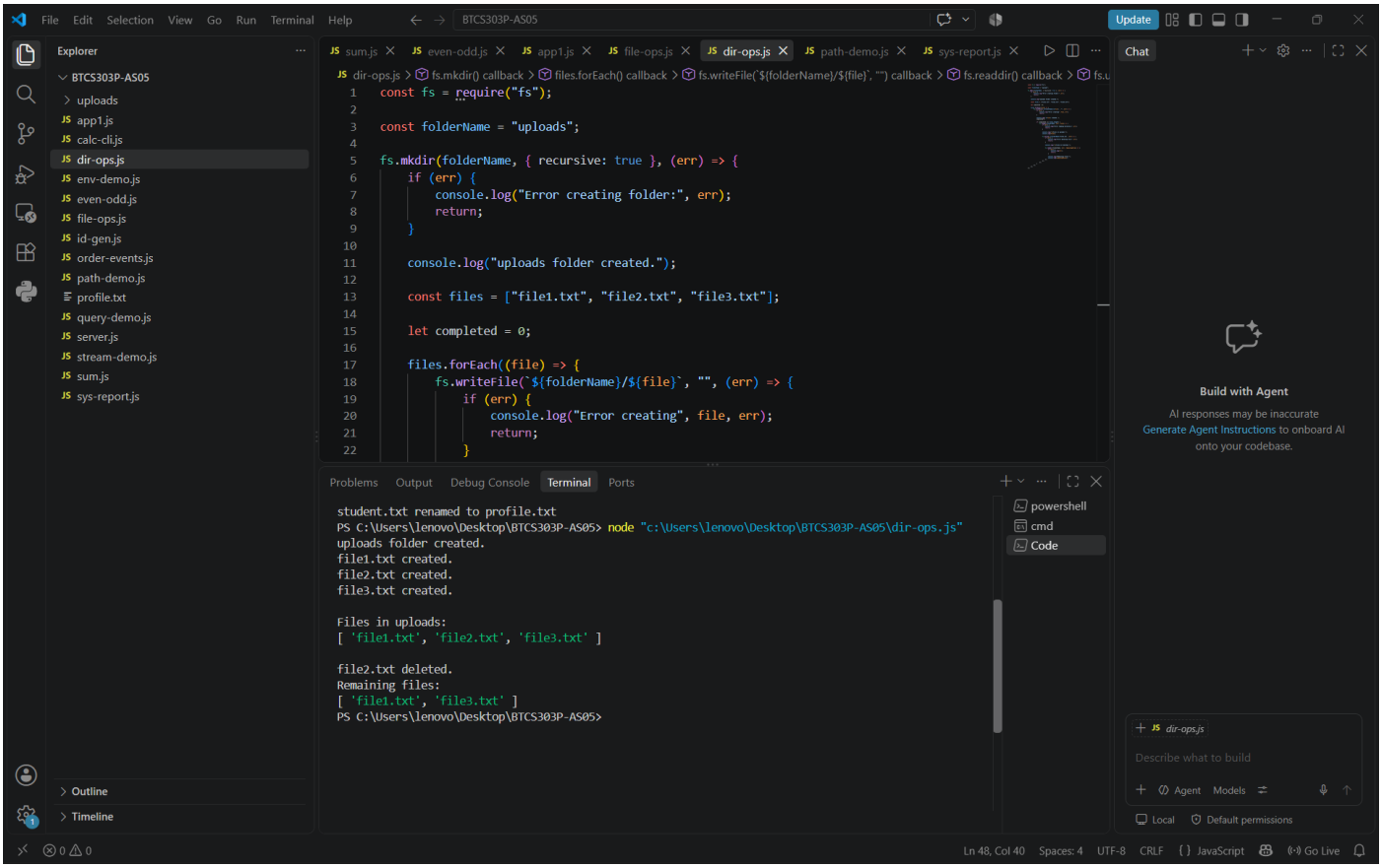
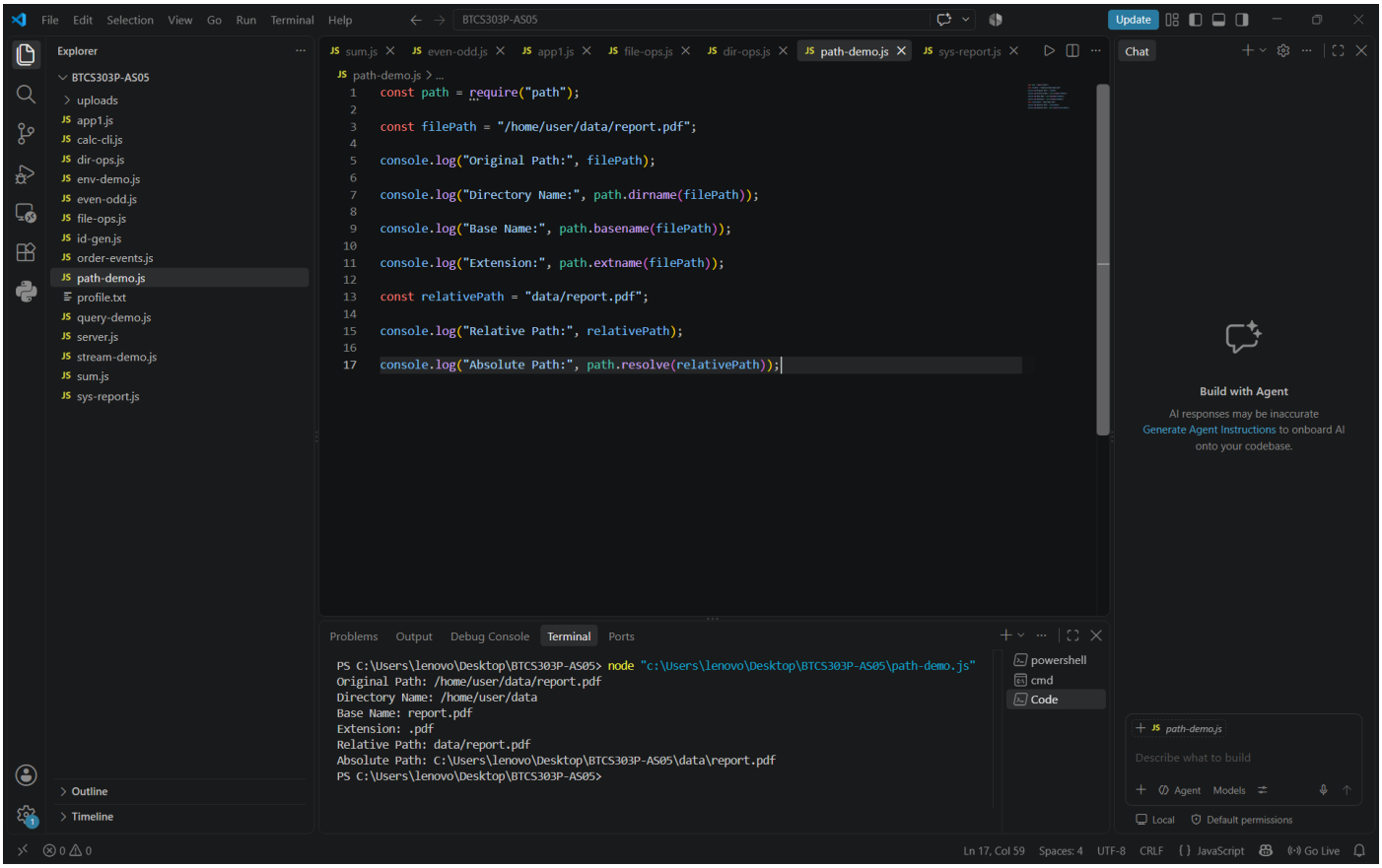
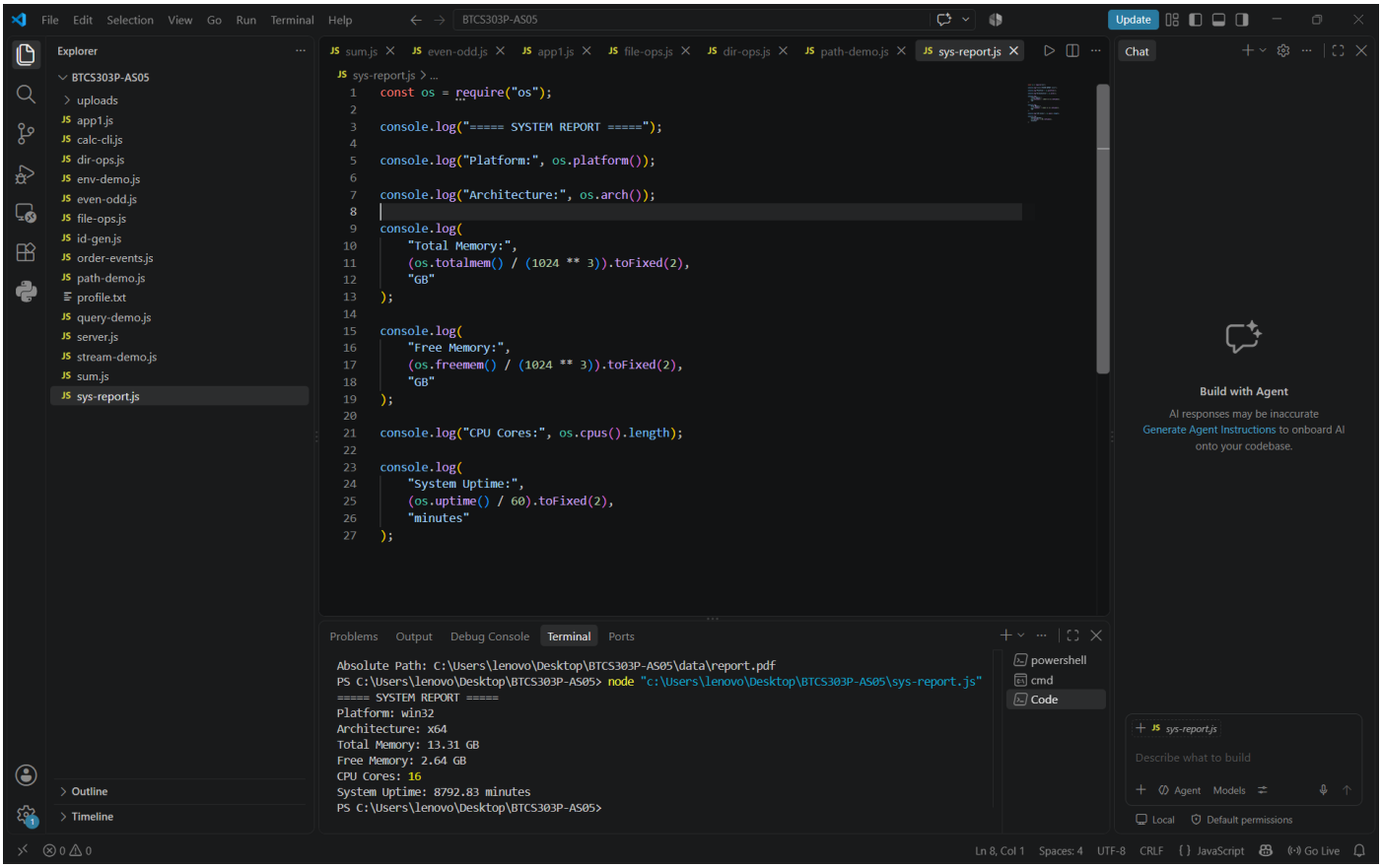

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app1.js
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path-demo.js
sys-report.js
app1.js > ...
1 const sum = require("./sum");
2 const evenOdd = require("./even-odd");
3
4 const a = 10;
5 const b = 20;
6
7 console.log("Sum:", sum(a, b));
8
9 const number = 15;
10
11 console.log("Number:", number);
12 console.log("Result:", evenOdd(number));
13
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems Output Debug Console Terminal Ports
PS C:\Users\lenovo\Desktop\BTCS303P-AS05> node "c:\Users\lenovo\Desktop\BTCS303P-AS05\app1.js"
Sum: 30
Number: 15
Result: Odd
PS C:\Users\lenovo\Desktop\BTCS303P-AS05>
powershell
cmd
Code
app1.js
Describe what to build
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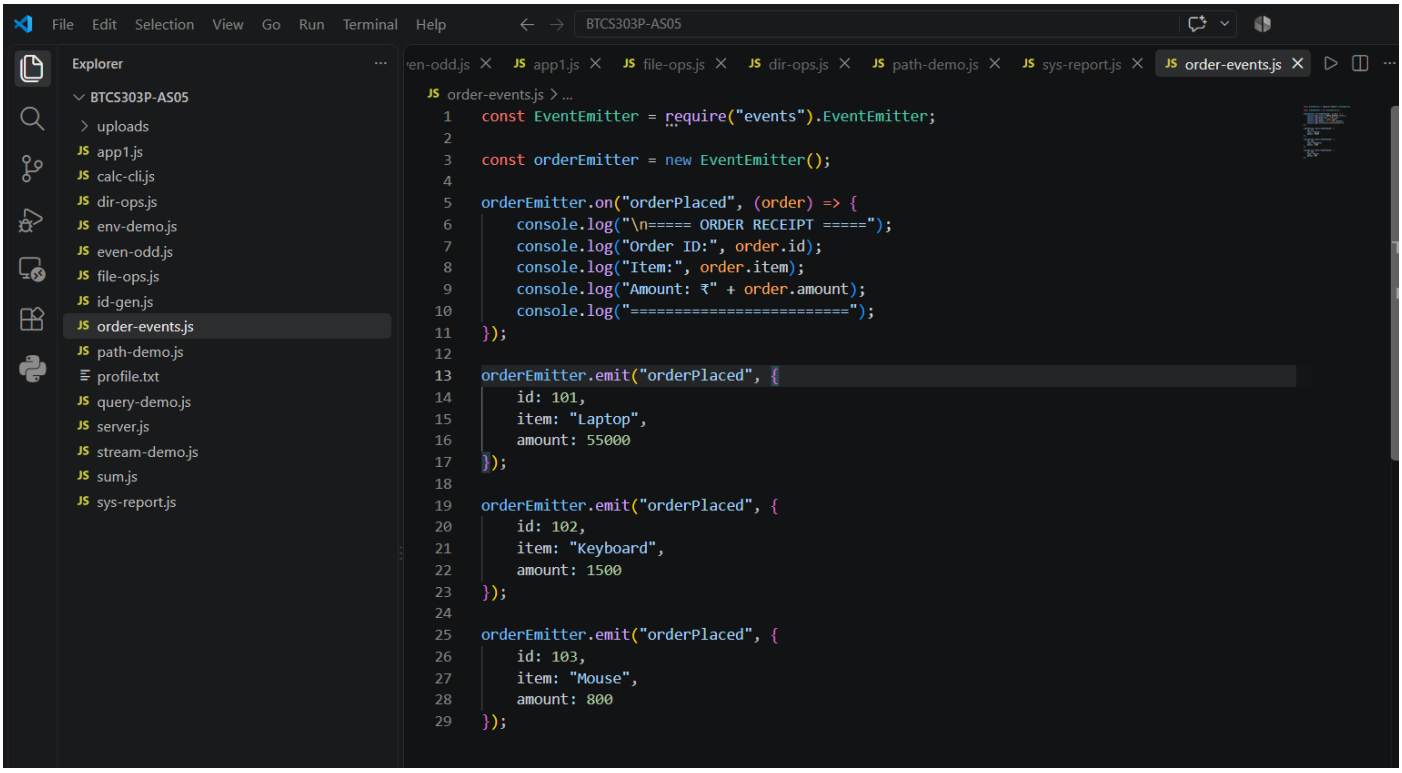

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sum.js
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app1.js
file-ops.js
dir-ops.js
path-demo.js
sys-report.js
file-ops.js > fs.writeFile("student.txt") callback > fs.appendFile("student.txt", "Course: Full Stack Web Development\n") callback
1 const fs = require("fs");
2
3 const studentData = `
4 Name: Ayush Sanwal
5 Roll Number: 243025065
6 `;
7
8 fs.writeFile("student.txt", studentData, (err) => {
9 if (err) {
10 console.log("Error writing file:", err);
11 return;
12 }
13
14 console.log("student.txt created and data written.");
15
16 fs.appendFile(
17 "student.txt",
18 "Course: Full Stack Web Development\n",
19 (err) => {
20 if (err) {
21 console.log("Error appending file:", err);
22 return;
23 }
24
25 console.log("Course name appended.");
26
27 fs.readFile("student.txt", "utf8", (err, data) => {
28 if (err) {
29 console.log("Error reading file:", err);
30 return;
31 }
32
33 console.log("\nFile Content:");
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems Output Debug Console Terminal Ports
File Content:
Name: Ayush Sanwal
Roll Number: 243025065
Course: Full Stack Web Development
student.txt renamed to profile.txt
PS C:\Users\lenovo\Desktop\BTCS303P-AS05>
powershell
cmd
Code
file-ops.js
Describe what to build
Agent Models
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Ln 21, Col 59 Spaces: 4 UTF-8 CRLF JavaScript Go Live

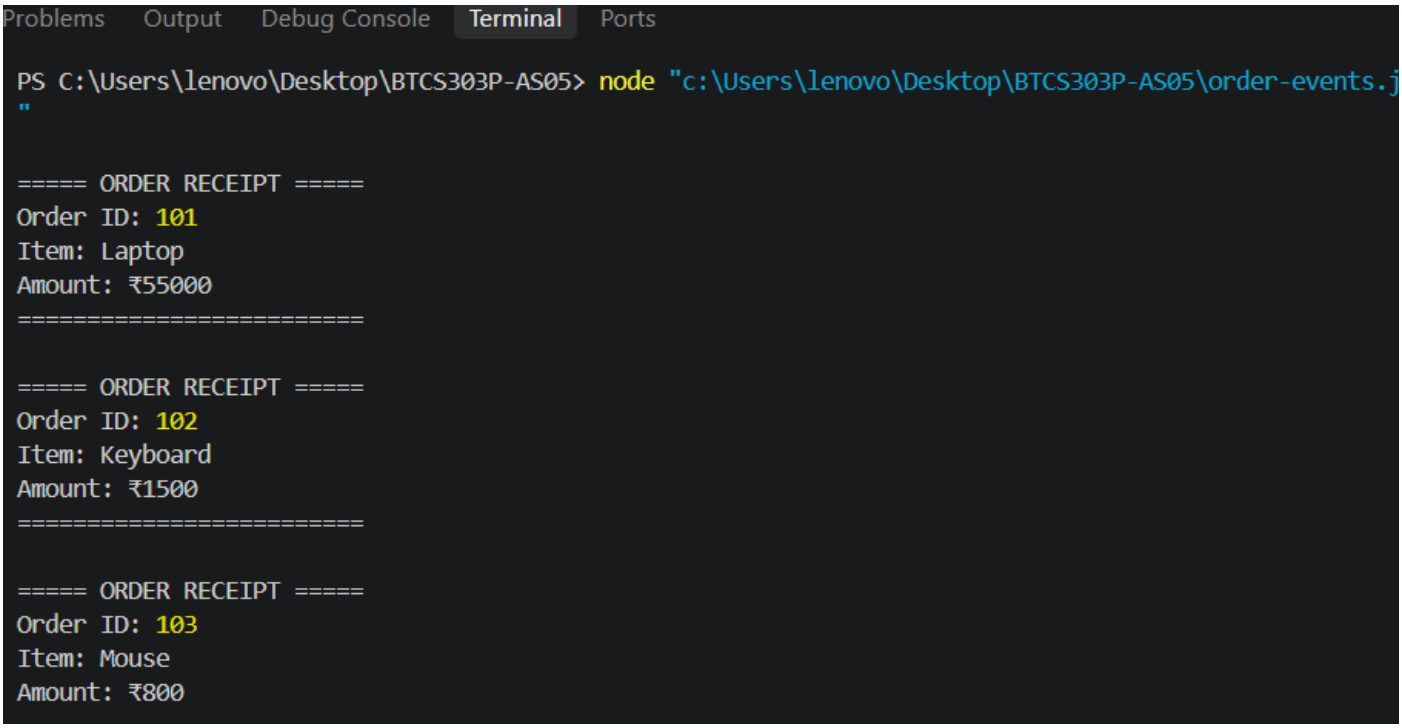

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sum.js
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file-ops.js
dir-ops.js
path-demo.js
sys-report.js
dir-ops.js > fs.mkdir() callback > files.forEach() callback > fs.writeFile(`${folderName}/${file}`, "") callback > fs.readdir() callback > fs.u
1 const fs = require("fs");
2
3 const folderName = "uploads";
4
5 fs.mkdir(folderName, { recursive: true }, (err) => {
6 if (err) {
7 console.log("Error creating folder:", err);
8 return;
9 }
10
11 console.log("uploads folder created.");
12
13 const files = ["file1.txt", "file2.txt", "file3.txt"];
14
15 let completed = 0;
16
17 files.forEach((file) => {
18 fs.writeFile(`${folderName}/${file}`, "", (err) => {
19 if (err) {
20 console.log("Error creating", file, err);
21 return;
22 }
23
24 console.log(`${file} created.`);
25 completed++;
26
27 if (completed === files.length) {
28 fs.readdir(folderName, (err, files) => {
29 if (err) {
30 console.log("Error reading directory:", err);
31 return;
32 }
33
34 console.log("\nFiles in uploads:");
35 console.log(files);
36
37 fs.unlink(`${folderName}/file2.txt`, (err) => {
38 if (err) {
39 console.log("Error deleting file:", err);
40 return;
Problems Output Debug Console Terminal Ports
file2.txt deleted.
powershell
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
dir-ops.js
Describe what to build
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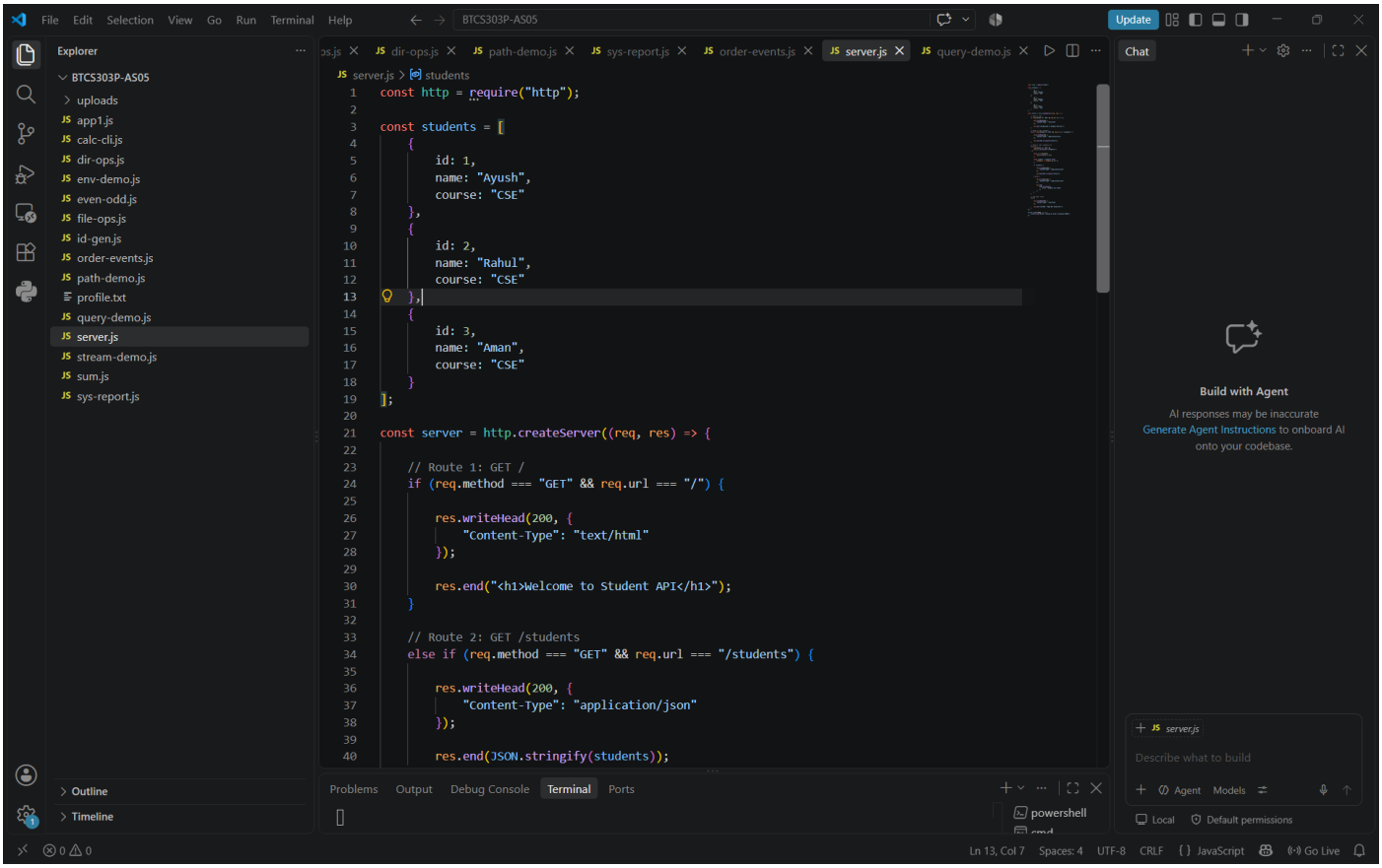

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sum.js
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dir-ops.js
path-demo.js
sys-report.js
dir-ops.js > fs.mkdir() callback > files.forEach() callback > fs.writeFile(`${folderName}/${file}`, "") callback > fs.readdir() callback > fs.u
1 const fs = require("fs");
2
3 const folderName = "uploads";
4
5 fs.mkdir(folderName, { recursive: true }, (err) => {
6 if (err) {
7 console.log("Error creating folder:", err);
8 return;
9 }
10
11 console.log("uploads folder created.");
12
13 const files = ["file1.txt", "file2.txt", "file3.txt"];
14
15 let completed = 0;
16
17 files.forEach((file) => {
18 fs.writeFile(`${folderName}/${file}`, "", (err) => {
19 if (err) {
20 console.log("Error creating", file, err);
21 return;
22 }
Problems Output Debug Console Terminal Ports
student.txt renamed to profile.txt
PS C:\Users\lenovo\Desktop\BTCS303P-AS05> node "c:\Users\lenovo\Desktop\BTCS303P-AS05\dir-ops.js"
uploads folder created.
file1.txt created.
file2.txt created.
file3.txt created.
Files in uploads:
['file1.txt', 'file2.txt', 'file3.txt']
file2.txt deleted.
Remaining files:
['file1.txt', 'file3.txt']
PS C:\Users\lenovo\Desktop\BTCS303P-AS05>
powershell
cmd
Code
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
dir-ops.js
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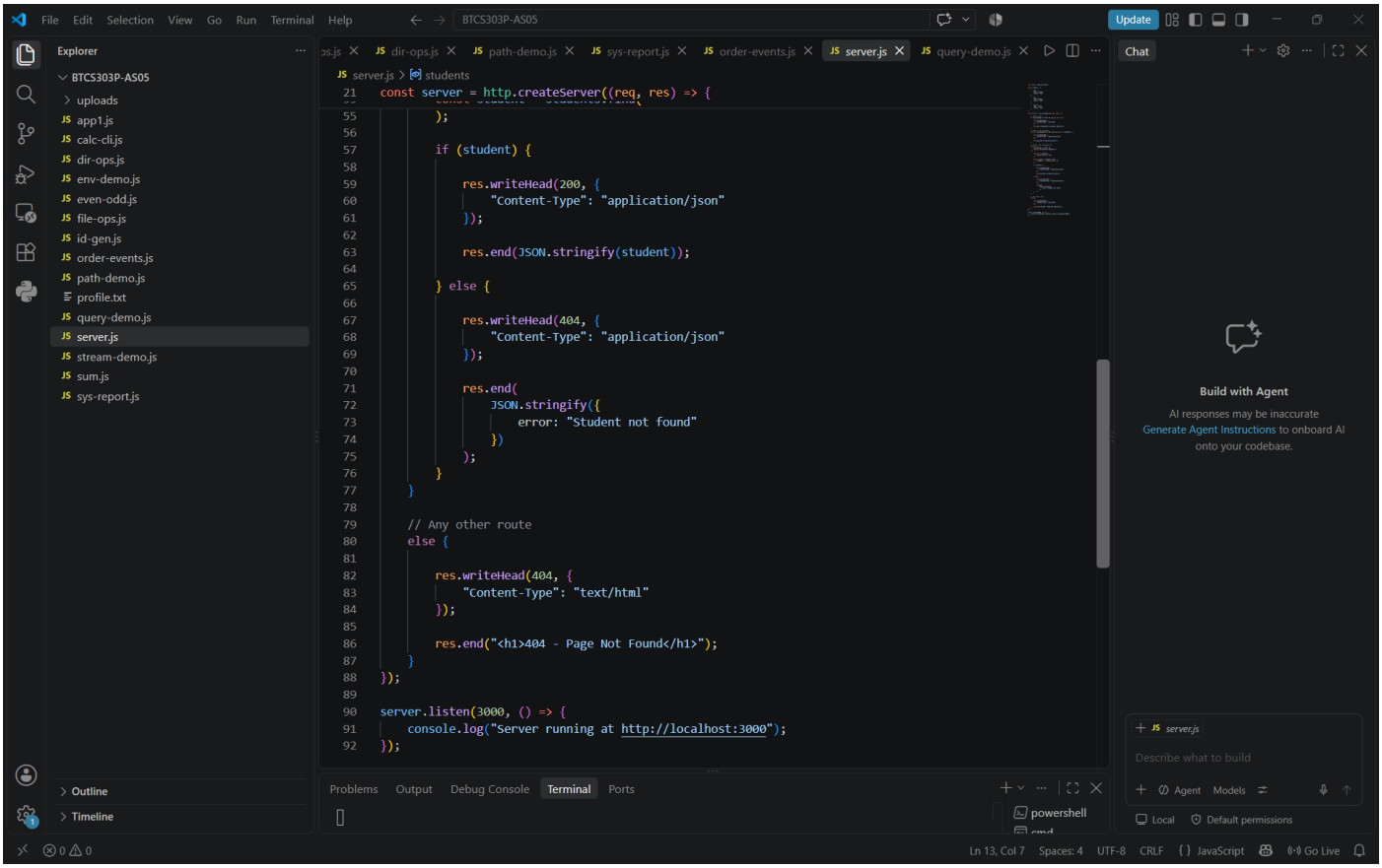

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sys-report.js
sum.js
even-odd.js
app1.js
file-ops.js
dir-ops.js
path-demo.js
sys-report.js
Chat
path-demo.js > ...
1 const path = require("path");
2
3 const filePath = "/home/user/data/report.pdf";
4
5 console.log("Original Path:", filePath);
6
7 console.log("Directory Name:", path.dirname(filePath));
8
9 console.log("Base Name:", path.basename(filePath));
10
11 console.log("Extension:", path.extname(filePath));
12
13 const relativePath = "data/report.pdf";
14
15 console.log("Relative Path:", relativePath);
16
17 console.log("Absolute Path:", path.resolve(relativePath));
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems Output Debug Console Terminal Ports
PS C:\Users\lenovo\Desktop\BTCS303P-AS05> node "C:\Users\lenovo\Desktop\BTCS303P-AS05\path-demo.js"
Original Path: /home/user/data/report.pdf
Directory Name: /home/user/data
Base Name: report.pdf
Extension: .pdf
Relative Path: data/report.pdf
Absolute Path: C:\Users\lenovo\Desktop\BTCS303P-AS05\data\report.pdf
PS C:\Users\lenovo\Desktop\BTCS303P-AS05>
powershell
cmd
Code
path-demo.js
Describe what to build
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sum.js
even-odd.js
app1.js
file-ops.js
dir-ops.js
path-demo.js
sys-report.js
Chat
sys-report.js > ...
1 const os = require("os");
2
3 console.log("===== SYSTEM REPORT =====");
4
5 console.log("Platform:", os.platform());
6
7 console.log("Architecture:", os.arch());
8
9 console.log(
10 "Total Memory:",
11 (os.totalmem() / (1024 ** 3)).toFixed(2),
12 "GB"
13);
14
15 console.log(
16 "Free Memory:",
17 (os.freemem() / (1024 ** 3)).toFixed(2),
18 "GB"
19);
20
21 console.log("CPU Cores:", os.cpus().length);
22
23 console.log(
24 "System Uptime:",
25 (os.uptime() / 60).toFixed(2),
26 "minutes"
27);
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems Output Debug Console Terminal Ports
Absolute Path: C:\Users\lenovo\Desktop\BTCS303P-AS05\data\report.pdf
PS C:\Users\lenovo\Desktop\BTCS303P-AS05> node "c:\Users\lenovo\Desktop\BTCS303P-AS05\sys-report.js"
===== SYSTEM REPORT =====
Platform: win32
Architecture: x64
Total Memory: 13.31 GB
Free Memory: 2.64 GB
CPU Cores: 16
System Uptime: 8792.83 minutes
PS C:\Users\lenovo\Desktop\BTCS303P-AS05>
powershell
cmd
Code
sys-report.js
Describe what to build
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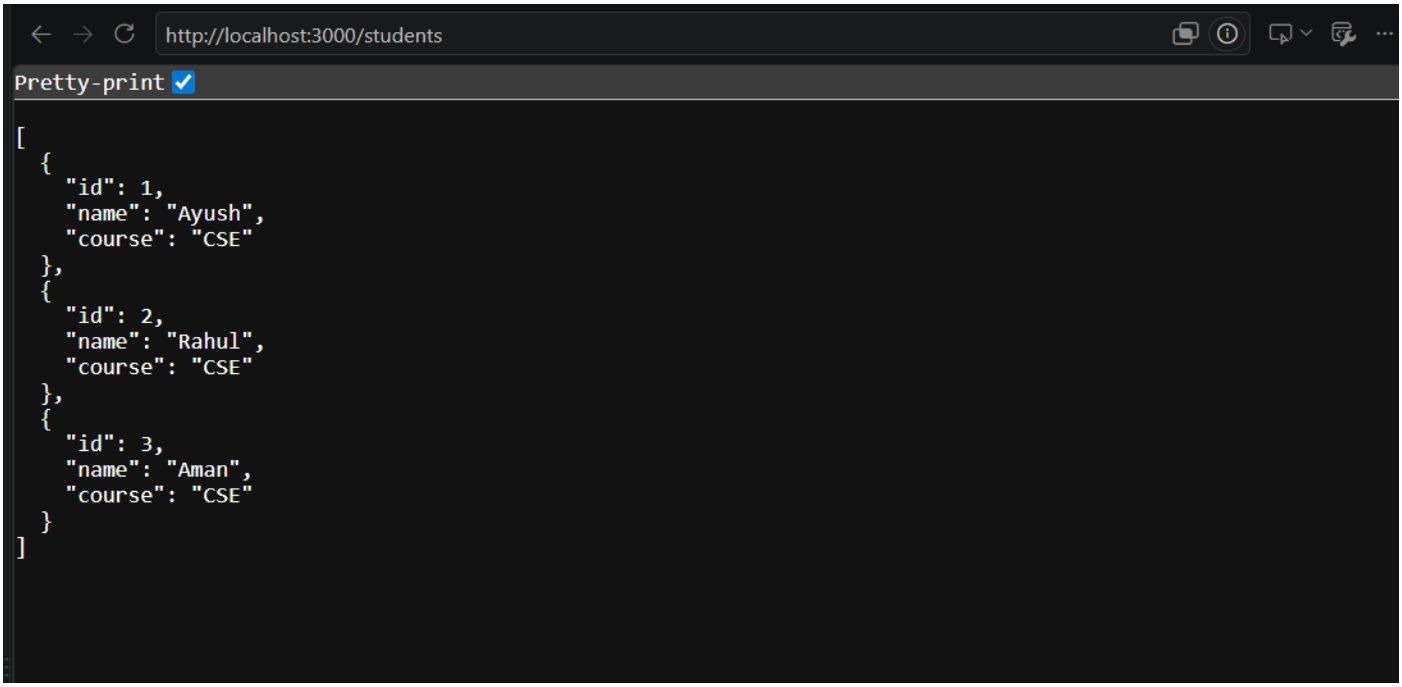

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stream-demo.js
sum.js
sys-report.js
app1.js
file-ops.js
dir-ops.js
path-demo.js
sys-report.js
order-events.js
order-events.js > ...
1 const EventEmitter = require("events").EventEmitter;
2
3 const orderEmitter = new EventEmitter();
4
5 orderEmitter.on("orderPlaced", (order) => {
6 console.log("\n===== ORDER RECEIPT =====");
7 console.log("Order ID:", order.id);
8 console.log("Item:", order.item);
9 console.log("Amount: ₹" + order.amount);
10 console.log("========================");
11 });
12
13 orderEmitter.emit("orderPlaced", {
14 id: 101,
15 item: "Laptop",
16 amount: 55000
17 });
18
19 orderEmitter.emit("orderPlaced", {
20 id: 102,
21 item: "Keyboard",
22 amount: 1500
23 });
24
25 orderEmitter.emit("orderPlaced", {
26 id: 103,
27 item: "Mouse",
28 amount: 800
29 });

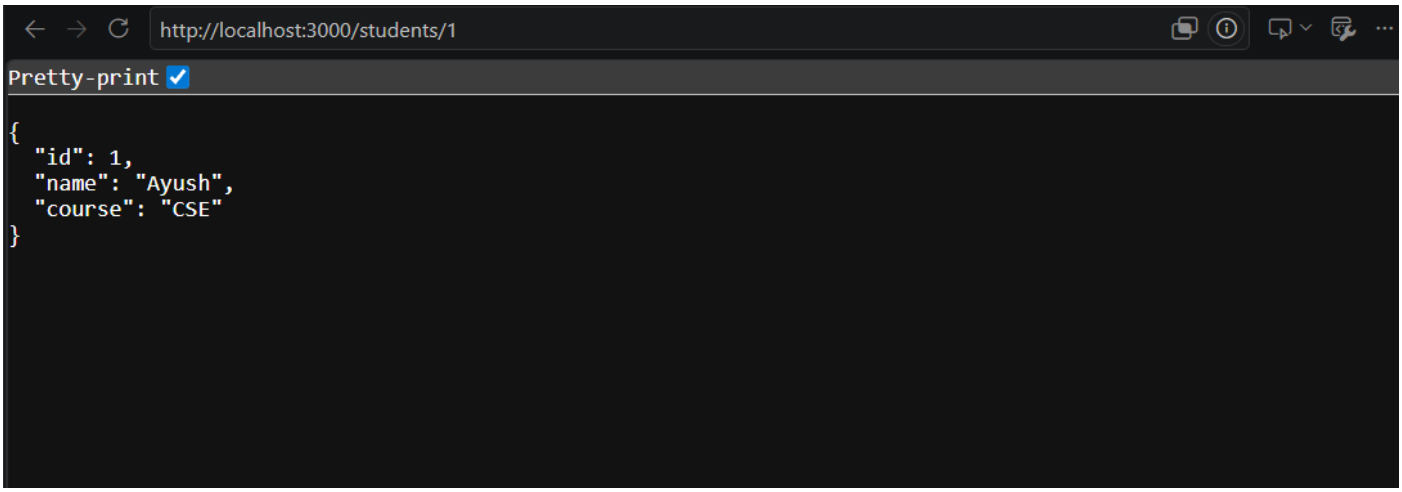

Problems Output Debug Console Terminal Ports
PS C:\Users\lenovo\Desktop\BTCS303P-AS05> node "c:\Users\lenovo\Desktop\BTCS303P-AS05\order-events.j
"
===== ORDER RECEIPT =====
Order ID: 101
Item: Laptop
Amount: ₹55000
========================
===== ORDER RECEIPT =====
Order ID: 102
Item: Keyboard
Amount: ₹1500
========================
===== ORDER RECEIPT =====
Order ID: 103
Item: Mouse
Amount: ₹800


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server.js
query-demo.js
server.js > students
const http = require("http");
const students = [
{
id: 1,
name: "Ayush",
course: "CSE"
},
{
id: 2,
name: "Rahul",
course: "CSE"
},
{
id: 3,
name: "Aman",
course: "CSE"
}
];
const server = http.createServer((req, res) => {
// Route 1: GET /
if (req.method === "GET" && req.url === "/") {
res.writeHead(200, {
"Content-Type": "text/html"
});
res.end("<h1>Welcome to Student API</h1>");
}
// Route 2: GET /students
else if (req.method === "GET" && req.url === "/students") {
res.writeHead(200, {
"Content-Type": "application/json"
});
res.end(JSON.stringify(students));
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dir-ops.js
path-demo.js
sys-report.js
order-events.js
server.js
query-demo.js
server.js > students
const server = http.createServer((req, res) => {
);
if (student) {
res.writeHead(200, {
"Content-Type": "application/json"
});
res.end(JSON.stringify(student));
} else {
res.writeHead(404, {
"Content-Type": "application/json"
});
res.end(
JSON.stringify({
error: "Student not found"
})
);
}
}
// Any other route
else {
res.writeHead(404, {
"Content-Type": "text/html"
});
res.end("<h1>404 - Page Not Found</h1>");
}
});
server.listen(3000, () => {
console.log("Server running at http://localhost:3000");
});
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server.js
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Welcome to Student API

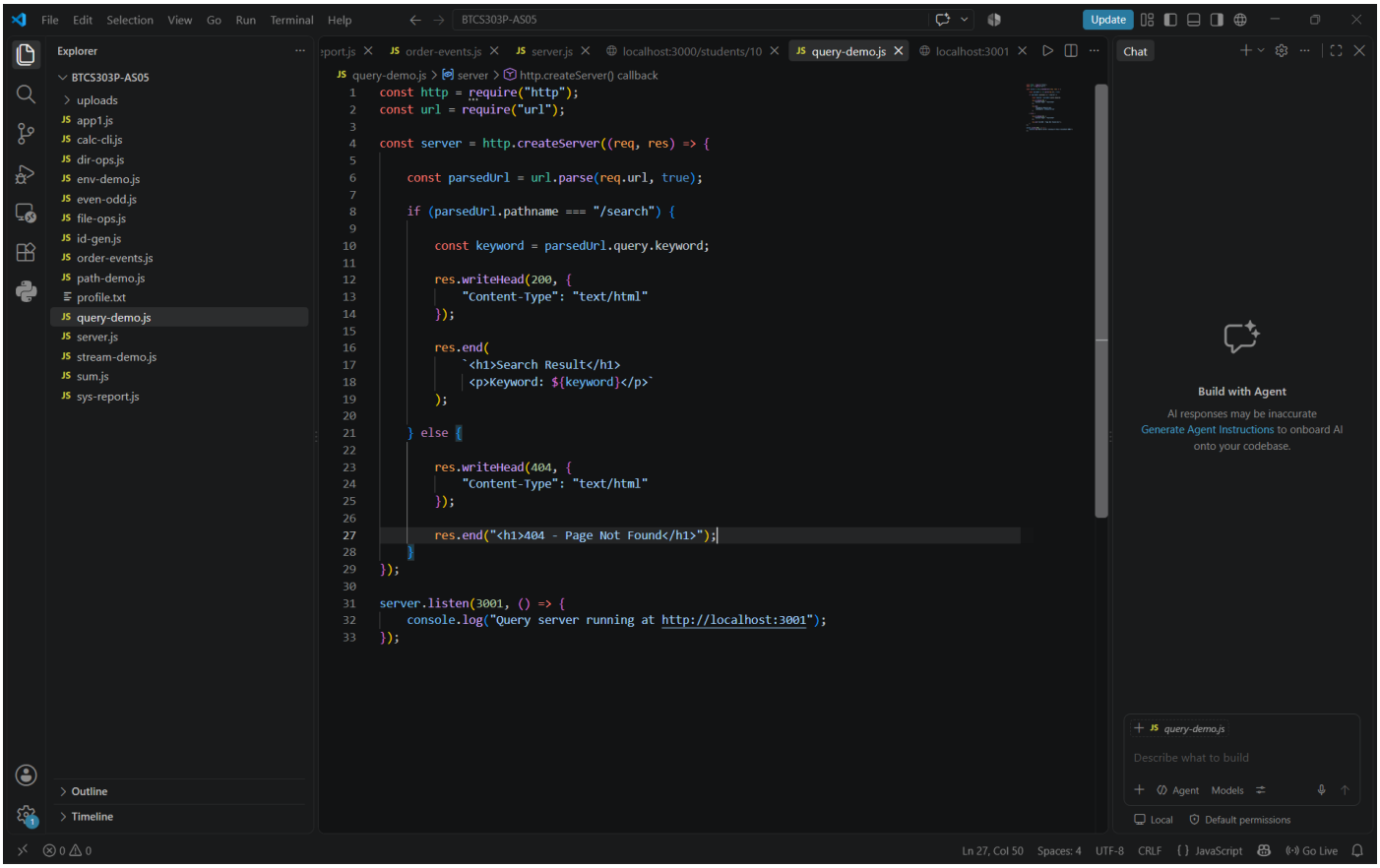


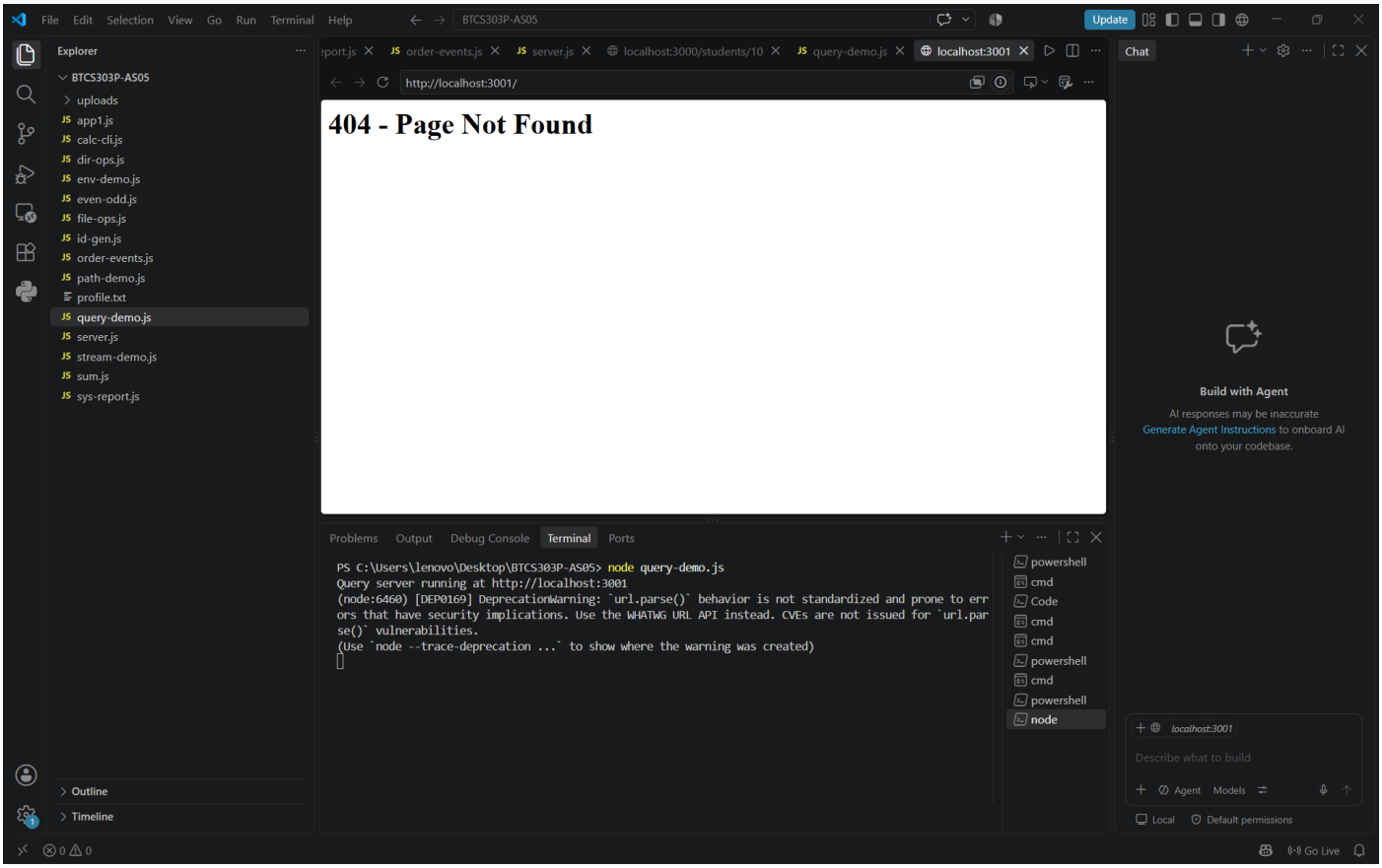


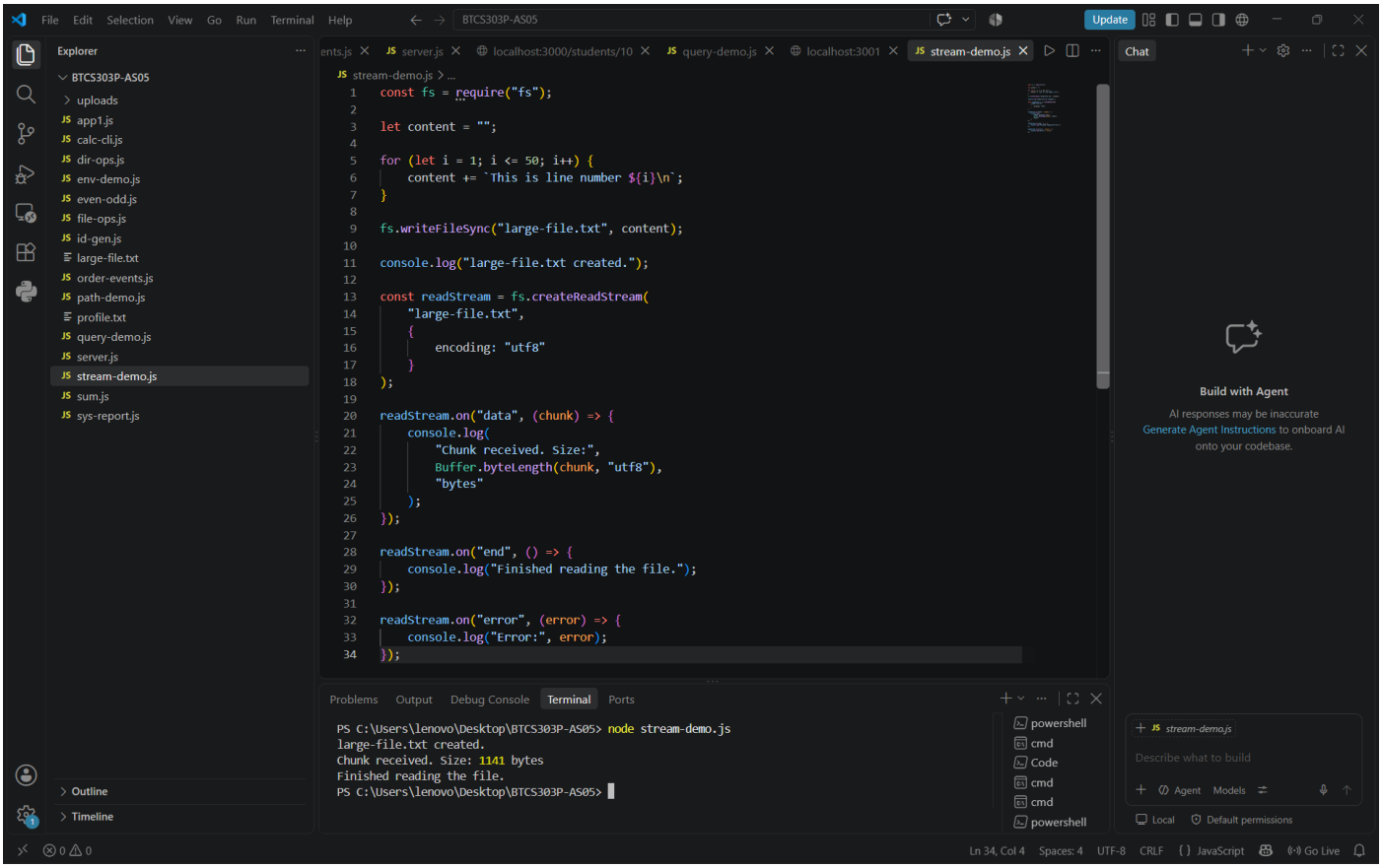
http://localhost:3000/students/10

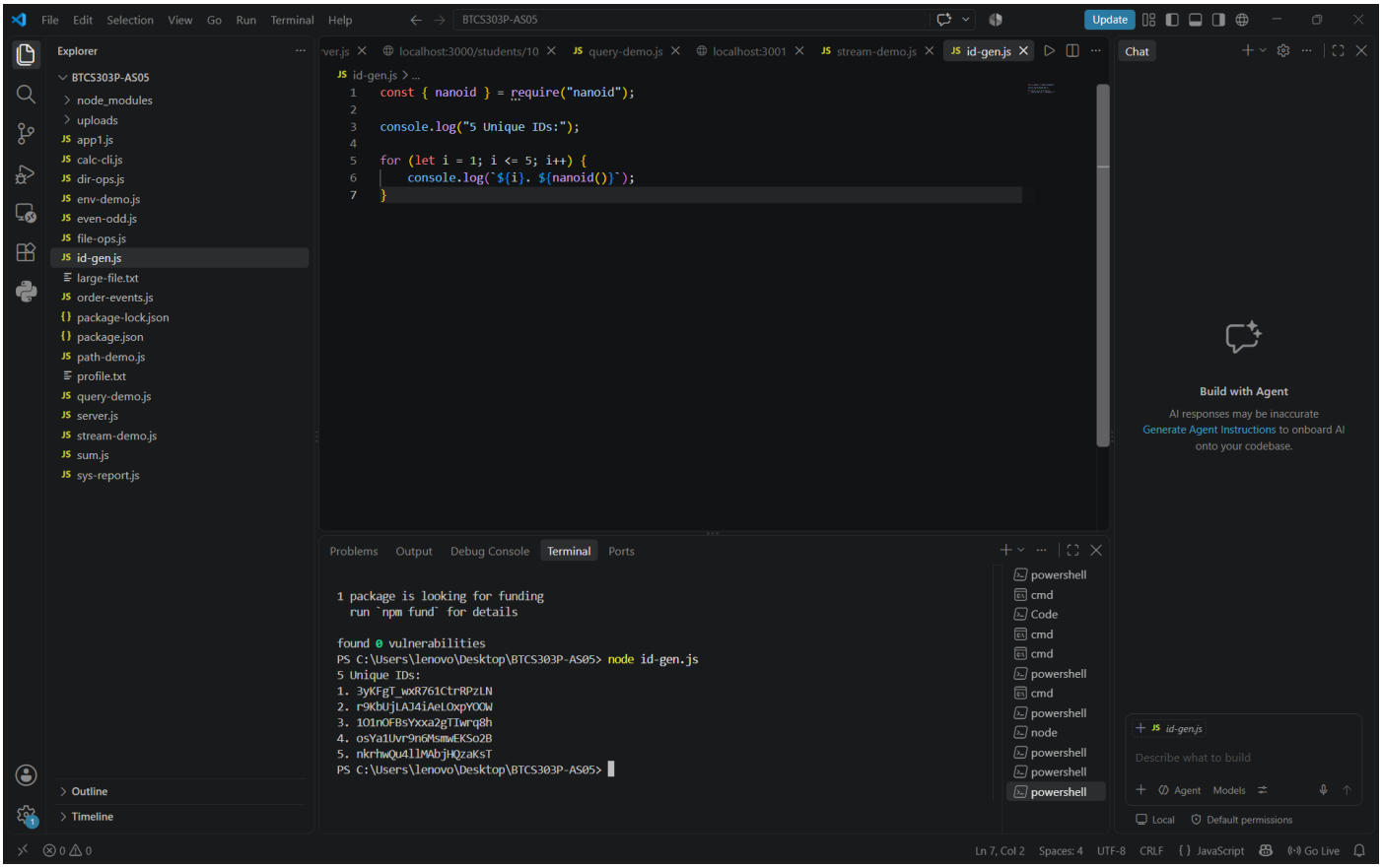
Pretty-print ☑

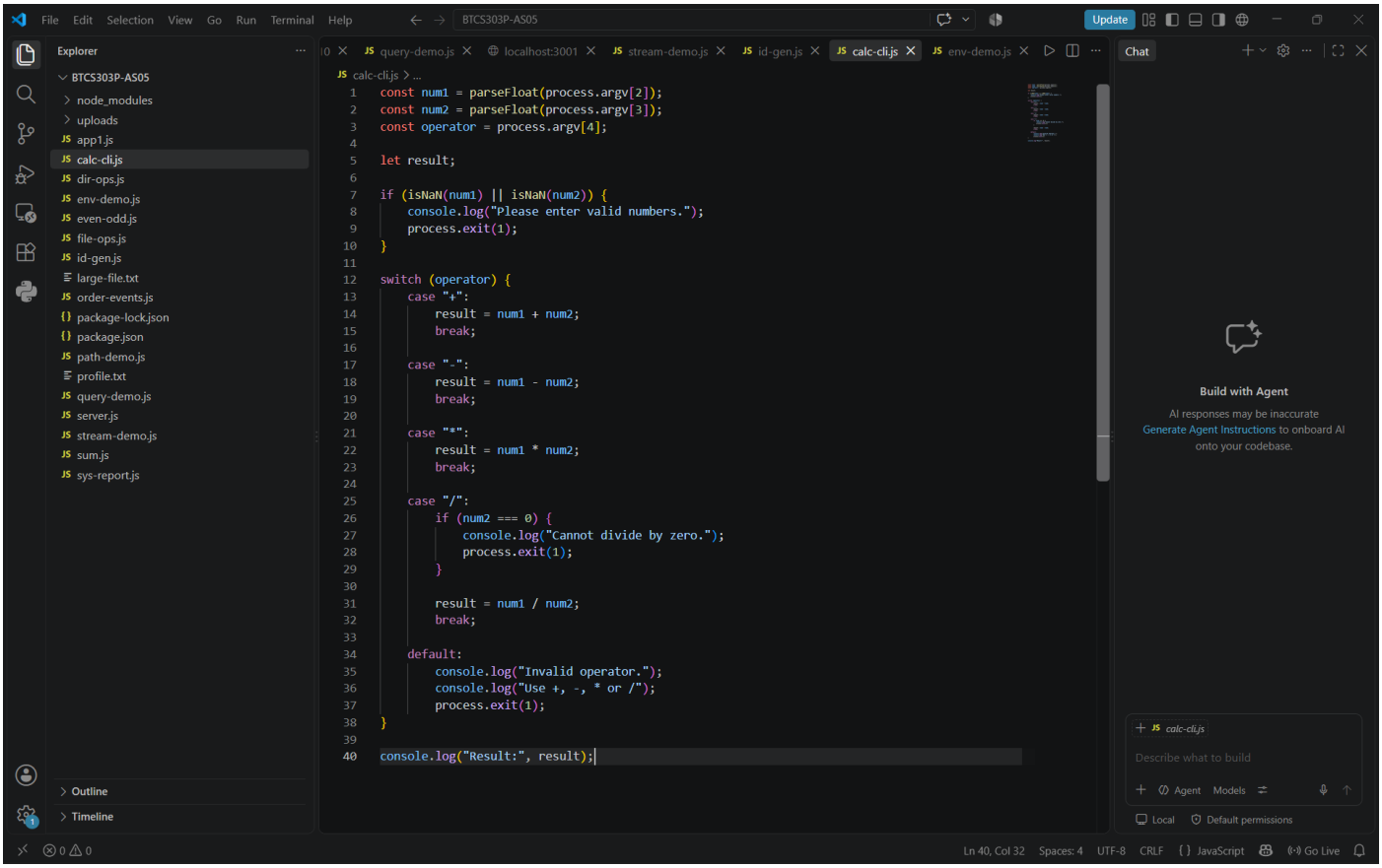

```
{
  "error": "Student not found"
}
```


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query-demo.js
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stream-demo.js
sum.js
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order-events.js
server.js
localhost:3000/students/10
query-demo.js
localhost:3001
http://localhost:3001/
404 - Page Not Found
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems Output Debug Console Terminal Ports
PS C:\Users\lenovo\Desktop\BTCS303P-AS05> node query-demo.js
Query server running at http://localhost:3001
(node:6460) [DEP0169] DeprecationWarning: `url.parse()` behavior is not standardized and prone to errors that have security implications. Use the WHATWG URL API instead. CVEs are not issued for `url.parse()` vulnerabilities.
(Use `node --trace-deprecation ...` to show where the warning was created)
powershell
cmd
Code
cmd
cmd
powershell
cmd
powershell
node
localhost:3001
Describe what to build
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Timeline
Go Live

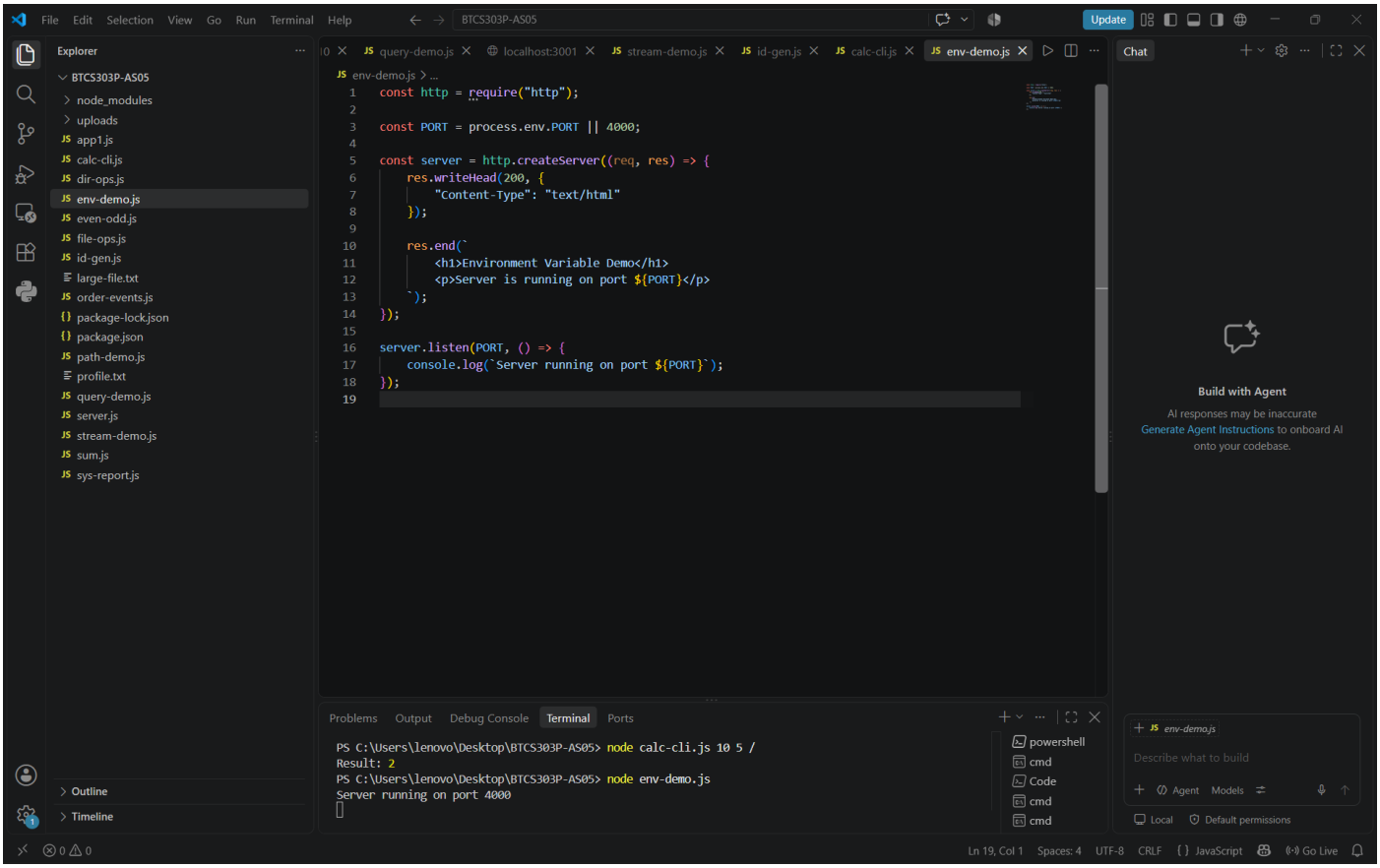

File Edit Selection View Go Run Terminal Help
BTCS303P-AS05
Update
Explorer
BTCS303P-AS05
uploads
app1.js
calc-cli.js
dir-ops.js
env-demo.js
even-odd.js
file-ops.js
id-gen.js
large-file.txt
order-events.js
path-demo.js
profile.txt
query-demo.js
server.js
stream-demo.js
sum.js
sys-report.js
server.js
localhost:3000/students/10
query-demo.js
localhost:3001
stream-demo.js
stream-demo.js > ...
1 const fs = require("fs");
2
3 let content = "";
4
5 for (let i = 1; i <= 50; i++) {
6 content += `This is line number ${i}\n`;
7 }
8
9 fs.writeFileSync("large-file.txt", content);
10
11 console.log("large-file.txt created.");
12
13 const readStream = fs.createReadStream(
14 "large-file.txt",
15 {
16 encoding: "utf8"
17 }
18);
19
20 readStream.on("data", (chunk) => {
21 console.log(
22 "Chunk received. Size:",
23 Buffer.byteLength(chunk, "utf8"),
24 "bytes"
25);
26 });
27
28 readStream.on("end", () => {
29 console.log("Finished reading the file.");
30 });
31
32 readStream.on("error", (error) => {
33 console.log("Error:", error);
34 });
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems Output Debug Console Terminal Ports
PS C:\Users\lenovo\Desktop\BTCS303P-AS05> node stream-demo.js
large-file.txt created.
Chunk received. Size: 1141 bytes
Finished reading the file.
PS C:\Users\lenovo\Desktop\BTCS303P-AS05>
powershell
cmd
Code
cmd
cmd
powershell
stream-demo.js
Describe what to build
Agent Models
Local Default permissions
Outline
Timeline
Ln 34, Col 4 Spaces: 4 UTF-8 CRLF JavaScript Go Live

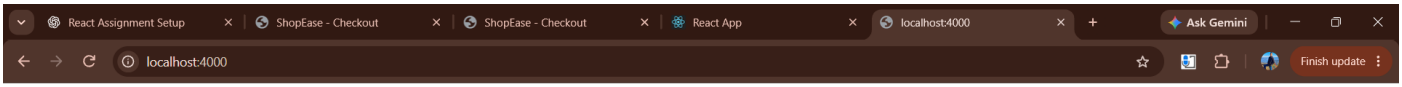

File Edit Selection View Go Run Terminal Help
BTCS303P-AS05
Update
Explorer
BTCS303P-AS05
node_modules
uploads
app1.js
calc-cli.js
dir-ops.js
env-demo.js
even-odd.js
file-ops.js
id-gen.js
large-file.txt
order-events.js
package-lock.json
package.json
path-demo.js
profile.txt
query-demo.js
server.js
stream-demo.js
sum.js
sys-report.js
Outline
Timeline
localhost:3000/students/10
query-demo.js
localhost:3001
stream-demo.js
id-gen.js
Chat
const { nanoid } = require("nanoid");
console.log("5 Unique IDs:");
for (let i = 1; i <= 5; i++) {
console.log(`${i}. ${nanoid()}`);
}
Problems Output Debug Console Terminal Ports
1 package is looking for funding
run `npm fund` for details
found 0 vulnerabilities
PS C:\Users\lenovo\Desktop\BTCS303P-AS05> node id-gen.js
5 Unique IDs:
1. 3yKFgT_wxR761CtrRPzLN
2. r9KbUjLAJ4iAeLOxpYOOW
3. 1O1nOFBsYxxa2gTIwrq8h
4. osYa1Uvr9n6MsmwEKSo2B
5. nkrhwQu4llMAbjHQzaKsT
PS C:\Users\lenovo\Desktop\BTCS303P-AS05>
powershell
cmd
Code
cmd
cmd
powershell
cmd
powershell
node
powershell
powershell
powershell
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
id-gen.js
Describe what to build
Agent Models
Local Default permissions
Ln 7, Col 2 Spaces: 4 UTF-8 CRLF JavaScript Go Live


File Edit Selection View Go Run Terminal Help
BTCS303P-AS05
Update
Explorer
BTCS303P-AS05
node_modules
uploads
app1.js
calc-cli.js
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even-odd.js
file-ops.js
id-gen.js
large-file.txt
order-events.js
package-lock.json
package.json
path-demo.js
profile.txt
query-demo.js
server.js
stream-demo.js
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sys-report.js
Outline
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query-demo.js
localhost:3001
stream-demo.js
id-gen.js
calc-cli.js
env-demo.js
Chat
const num1 = parseFloat(process.argv[2]);
const num2 = parseFloat(process.argv[3]);
const operator = process.argv[4];
let result;
if (isNaN(num1) || isNaN(num2)) {
console.log("Please enter valid numbers.");
process.exit(1);
}
switch (operator) {
case "+":
result = num1 + num2;
break;
case "-":
result = num1 - num2;
break;
case "*":
result = num1 * num2;
break;
case "/":
if (num2 === 0) {
console.log("Cannot divide by zero.");
process.exit(1);
}
result = num1 / num2;
break;
default:
console.log("Invalid operator.");
console.log("Use +, -, * or /");
process.exit(1);
}
console.log("Result:", result);
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
calc-cli.js
Describe what to build
Agent Models
Local Default permissions
Ln 40, Col 32 Spaces: 4 UTF-8 CRLF JavaScript Go Live

```
Path
----
C:\Users\lenovo\Desktop\BTCS303P-AS05


PS C:\Users\lenovo\Desktop\BTCS303P-AS05> node calc-cli.js 10 5 +^B
Invalid operator.
Use +, -, * or /
PS C:\Users\lenovo\Desktop\BTCS303P-AS05> node calc-cli.js 10 5 +
Result: 15
PS C:\Users\lenovo\Desktop\BTCS303P-AS05> node calc-cli.js 10 5 -
Result: 5
PS C:\Users\lenovo\Desktop\BTCS303P-AS05> node calc-cli.js 10 5 "*"
Result: 50
PS C:\Users\lenovo\Desktop\BTCS303P-AS05> node calc-cli.js 10 5 /
Result: 2
PS C:\Users\lenovo\Desktop\BTCS303P-AS05>
```





Environment Variable Demo

Server is running on port 4000
